

SUPPLEMENT TO “WHO CHOOSES AND WHO BENEFITS? THE DESIGN OF PUBLIC  
SCHOOL CHOICE SYSTEMS”

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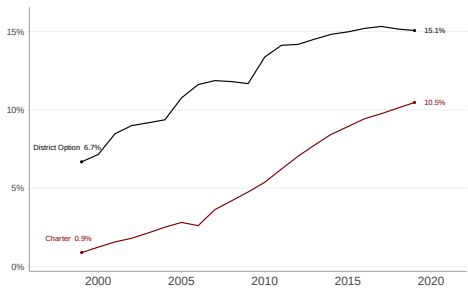
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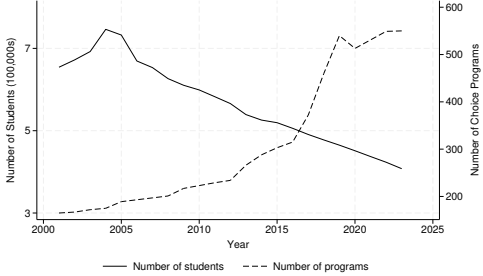
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APPENDIX A: APPENDIX FIGURES AND TABLES



(a) Public School Choice and Charter School Enrollment Trends (150 Largest Districts)



(b) LAUSD Enrollment and Choice Availability

FIGURE A.1.—Choice-School Enrollment and Availability Trends. *Notes:* Panel (a) reports enrollment shares for charter schools and district-operated choice schools (“District Option”) for the 150 large school districts. Data are from the Common Core of Data. For a given district, a local charter includes all charter schools that have the district’s Local Education Agency ID or operate in a ZIP code served by a traditional public school with that district’s LEAID. To identify district-operated choice schools, we use the Common Core’s magnet flag, since other forms of choice are not reliably recorded. In a few states (e.g., Ohio), magnet status is not reported. For that reason, we also count a school as a within-district choice option if the word “magnet” appears in its name. The district-operated choice numbers in Panel (a) are potentially conservative, since many districts have choice options (e.g., gifted and talented schools) that are not officially named as magnets and otherwise are not easily identified. Panel (b) reports trends in LAUSD district-level enrollment and in the availability of LAUSD choice programs. The solid line shows LAUSD enrollment in Grades 1–12 between 2001 and 2023. The dashed line shows the total number of choice programs over the same period, where choice programs include magnet programs, dual-language programs, and affiliated charter schools.

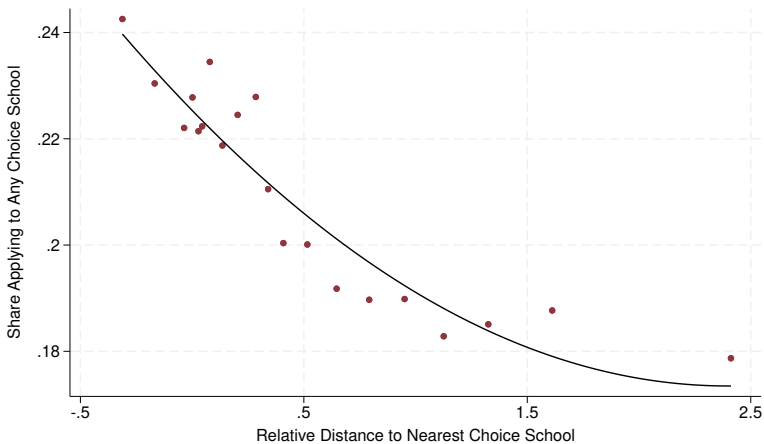


FIGURE A.2.—School Distance Predicts Participation in the Choice System. *Note:* This figure reports a binscatter plot assessing the relationship between students’ likelihood of applying to choice programs and relative distance to the nearest choice programs within LAUSD subdistricts. There are six LAUSD subdistricts that segment the district into large areas. Choice programs include magnet programs, dual-language programs, and district-run affiliated charter schools. The sample comprises student-level data for all cohorts of fifth-grade students from 2004 to 2017.

FIGURE A.3.—Expansions in Access to LAUSD Choice Schools

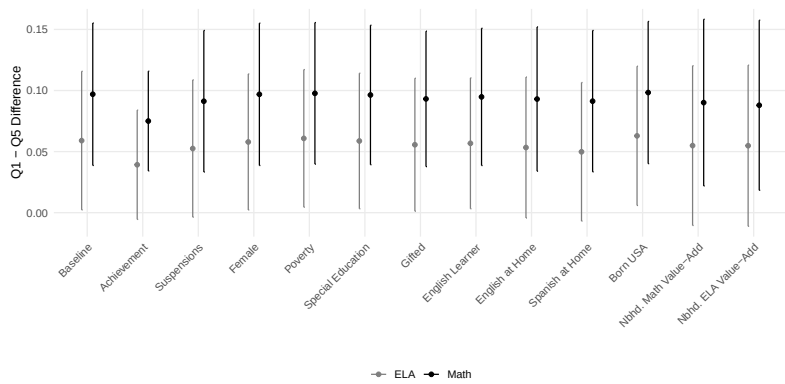
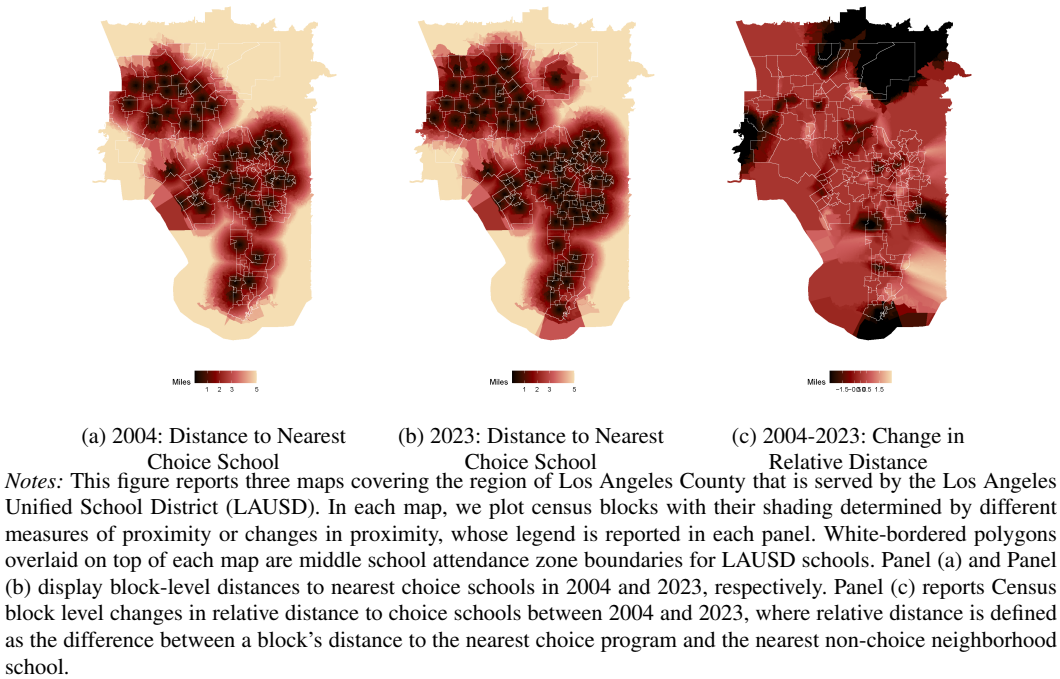
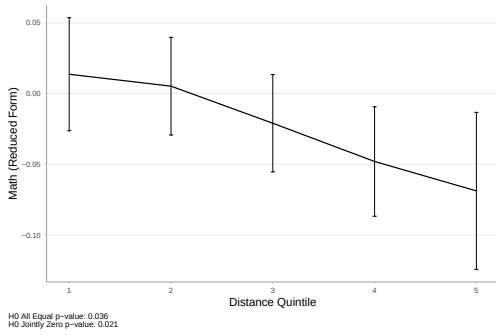
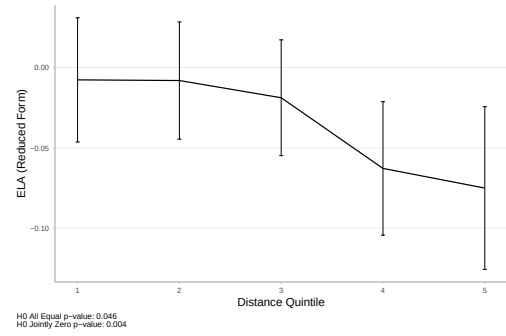


FIGURE A.4.—Robustness of Distance Gradient in Achievement Impacts. *Note:* This figure assesses the robustness of the estimated distance gradient: the difference in the estimated impacts of winning a lottery for students who live closest to and furthest from their nearest choice school. All estimates are based on Equation (2) in the main paper, where the coefficients of interest are interactions between a lottery offer indicator and indicators for distance quintiles based on students' residential addresses and distance to choice schools. The leftmost results correspond to our baseline specifications for ELA and math achievement, respectively, as identified in the figure legend. Each additional pair of estimates augments the baseline model with a main effect for the corresponding covariate—for example, baseline achievement or neighborhood-school value added—and an interaction between that covariate and the offer indicator. We report the estimated difference in reduced-form effects between quintiles 1 (Q1) and 5 (Q5), together with its 95 percent confidence interval. Standard errors are clustered at the lottery-strata level.

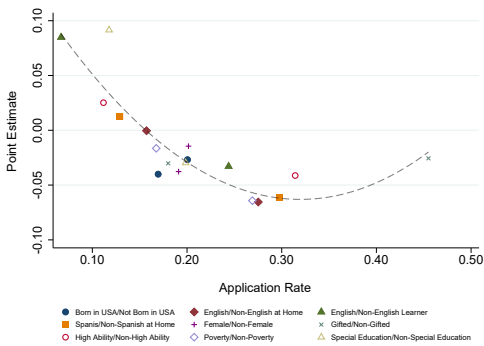


(a) Math

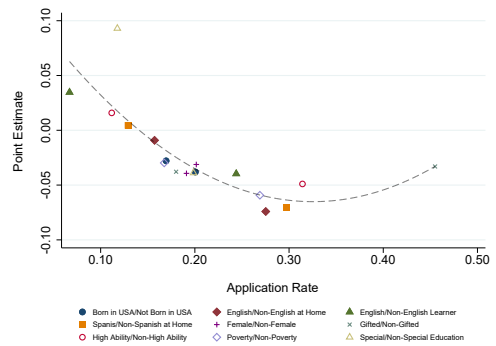


(b) ELA

FIGURE A.5.—Lottery Effects by Similarity Index. *Note:* This figure reports reduced form effects of receiving an offer on Math and ELA achievement in Panel (a) and Panel (b), respectively. We report estimates for different quintiles of a similarity index constructed using the observable characteristics in Figure A.4. Specifically, we standardize each covariate to have a mean of zero and a unit variance, and then calculate the Euclidean distance of each student to the average applicant to that choice program in that same year. The estimates come from a regression of achievement—Math or ELA—on lottery strata fixed effects, main effects for preference index quintiles, and interactions of offers and preference quintiles. We report estimates of the interactions in each sub-figure. We report 95-percent confidence intervals that use robust standard errors clustered at the lottery strata level. We also report  $p$ -values from hypothesis tests that test the null hypothesis that the preference-index effects are equal and another test that tests the null hypothesis that the preference-index effects are jointly equal to zero.



(a) Math



(b) ELA

FIGURE A.6.—Subgroup Heterogeneity in Application Rates and Lottery Effects. *Note:* This figure reports subgroup heterogeneity in lottery-based treatment effects and application rates. As indicated in the legend, each marker type represents results for a given student subgroup and its complement (e.g., English versus Non-English Learner students). With the exception of baseline achievement, all subgroups are based on binary pre-determined characteristics. In the case of achievement, we define high and low-ability sub-groups as those above and below the median of baseline achievement. We omit creating a subgroup based on suspensions, as it is a relatively rare outcome. The  $y$ -axis reports the given subgroups treatment effect (e.g., the effect of receiving an offer to attend a choice school for English Learners). All estimates are from separate subgroup specific models where we regress an achievement outcome on the lottery offer  $Z_i$  and lottery fixed effect. The  $x$ -axis reports the corresponding choice-school application rate for a given subgroup, which we measure as the share of all 5th graders who submitted at least one choice school application over our sample period. Panels (a) and (b) report results for Math and ELA, respectively. Detailed estimates are included in Appendix Table A.5.

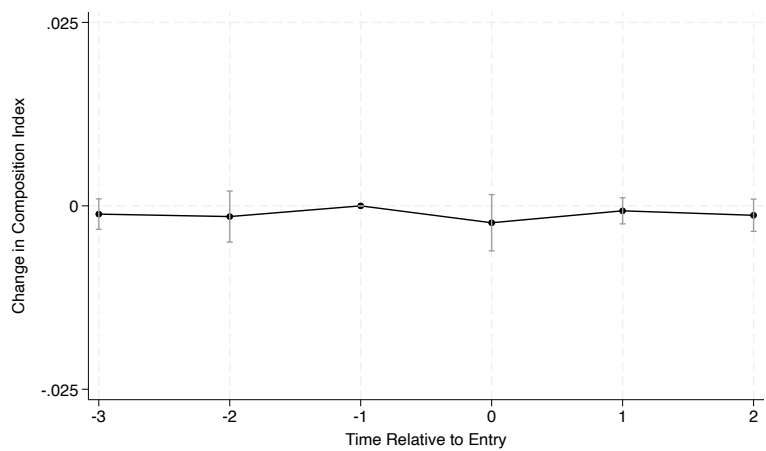
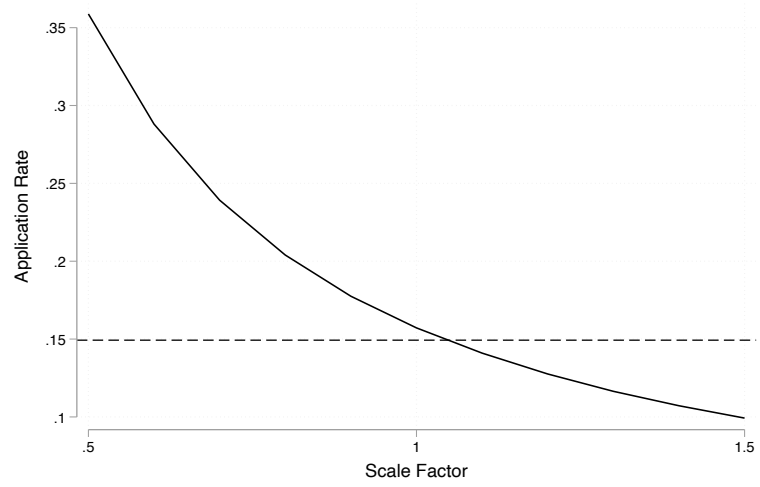
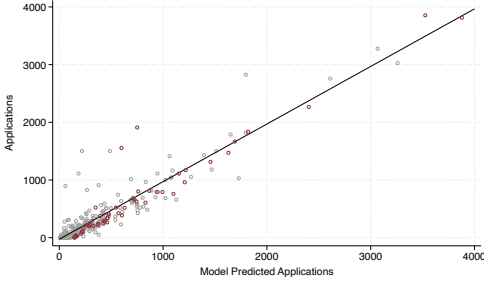


FIGURE A.7.—Changes in Student Composition in Response to Choice School Entry. *Note:* This figure reports event-study estimates based on Equation (1) in the main paper, where the dependent variable is a neighborhood-level composition index among applicants to choice schools. The composition index is constructed by first calculating fitted values from a regression of student-level achievement on a vector of baseline covariates and then averaging those fitted values among applicants within a candidate neighborhood. We report 95 percent confidence intervals based on robust standard errors clustered at the neighborhood level.

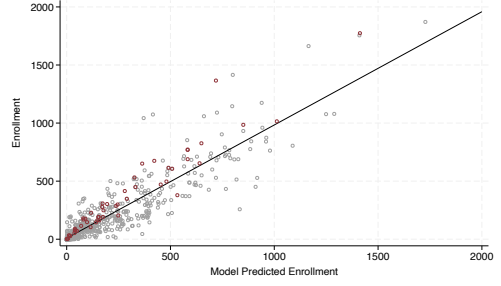
FIGURE A.8.—Application Cost Identification Exercise



*Notes:* This figure reports results from an identification exercise for the application costs in the model specified in Section 7.2 of the main paper. The vertical axis corresponds to application rates in the simulated data while the horizontal axis varies estimated application costs ( $c(a, X_i, \eta_i)$ ), where we scale costs by corresponding number on the horizontal axis. The black line reports the corresponding application rate as we scale costs while holding other parameters of the model constant. For example, at scale factor equal to one, the value of the black line corresponds to the simulated application rate when cost parameters are evaluated at the values that maximize the likelihood. Similarly, at scale factor equal to 0.5, we scale costs by 0.5 and hold other parameters of the model constant at the values that maximize the likelihood. The dashed black line corresponds to application rates observed in the data. The intuition is that a moment is important for a parameter’s identification if, as we move across the scaling factor axis, the values of simulated application rates change and cross the horizontal dashed line (i.e., the value of that moment in the real-world data). Overall, this exercise provides graphical evidence of how application costs are identified.

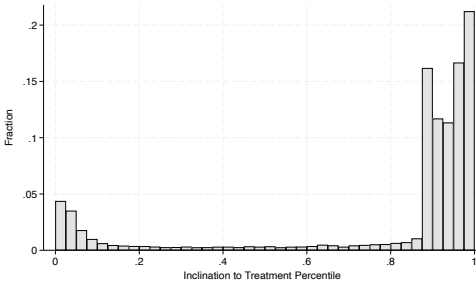
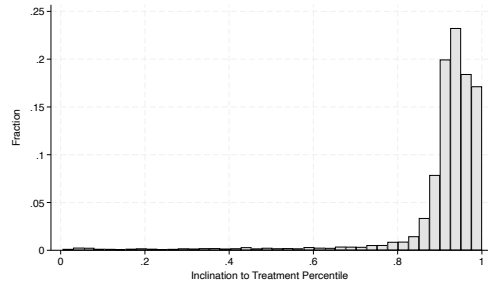


(a) Applications



(b) Enrollment

FIGURE A.9.—Model Fit: Application and Enrollment. *Note:* This figure reports an assessment of the overall fit of the model based on out-of-sample validation. We estimate our preferred model on a 25% random training sample and assess its fit on the remaining 75%. We run 200 simulations in which we use parameter estimates from the training sample to simulate applications, offers, and enrollment in the hold-out sample. The figure plots school-level aggregates based on simulated averages and the actual data. Panel (a) reports school-level observed number of applications in the holdout sample ( $y$ -axis) against the model-predicted applications in the holdout sample ( $x$ -axis). Maroon markers correspond to schools, or in other words, assess the fit of school-level statistics, while the grey markers further interact school with student observable attributes. Panel (b) reports similar statistics for enrollment.

(a)  $\theta$  Percentile

(b) Preference Index Percentile

FIGURE A.10.—Distribution of Lottery Applicant Choice School Preferences. *Note:* This figure focuses on the sample of students who are in the lottery estimation sample (i.e., students who applied to oversubscribed choice schools) and reports the distribution of their estimated posterior means of  $\theta_i$  and a preference index measure in Panels (a) and (b), respectively. The preference index is defined as  $\mathcal{I}_i \equiv \beta_c X_i + \theta_i$ , which can be interpreted as student  $i$ 's utility from enrolling in a choice school as a function of both observables and unobservables. Both figures demonstrate that students that are actively engaging in the choice process are drawn from regions of the distribution that are associated with the smallest learning gains.

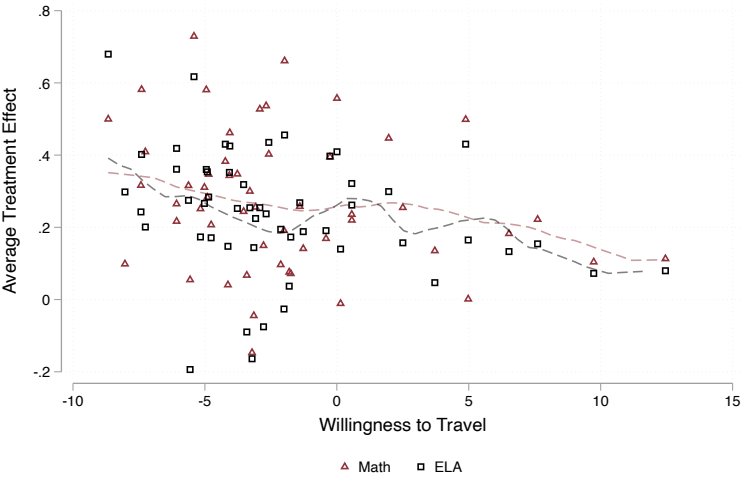
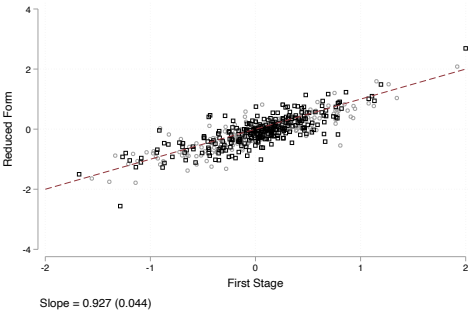
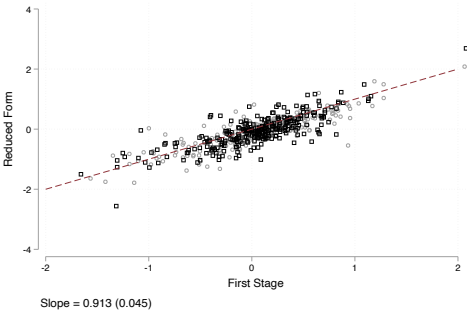


FIGURE A.11.—Demand for Effective Choice Schools. *Note:* This figure reports the relationship between estimated school effectiveness and demand, where demand is measured in miles of willingness to travel. School effectiveness is estimated in a model that adjusts for selection into application and enrollment using posterior-mean control-function estimates from the estimated demand model.



(a) Student Achievement Model 1



(b) Student Achievement Model 2

FIGURE A.12.—Model Validation: Visual Instrumental Variables Plot. *Note:* This figure reports results from a validation of our student achievement outcome model. Each dot in the figure represents results from an oversubscribed choice school lottery (specifically, the  $N = 420$  lotteries that overlap with the 2004–2013 time window used to estimate the structural model). The “first-stage” ( $x$ -axis) is the estimated effect of a randomized offer  $Z_i$  in the given lottery on the model-based prediction of student achievement based on the model from Section 7.3 of the main paper. The “reduced form” ( $y$ -axis) is the estimated effect of the same randomized offer  $Z_i$  in the given lottery on observed achievement. Panels (a) and (b) report results from the type-probability and linear control function specifications, respectively. The slope (i.e., the IV “forecast coefficient”) equals 0.91 (0.06), so we cannot reject 1, indicating the outcome model predicts causal impacts up to sampling error. All specifications include lottery-stratum fixed effects.

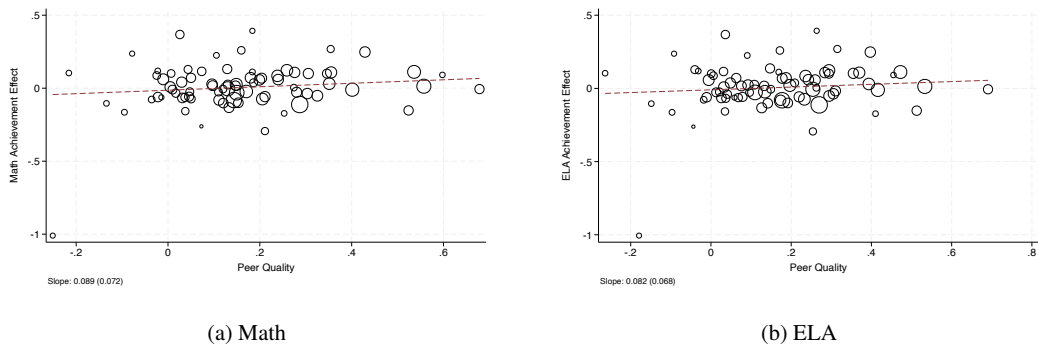
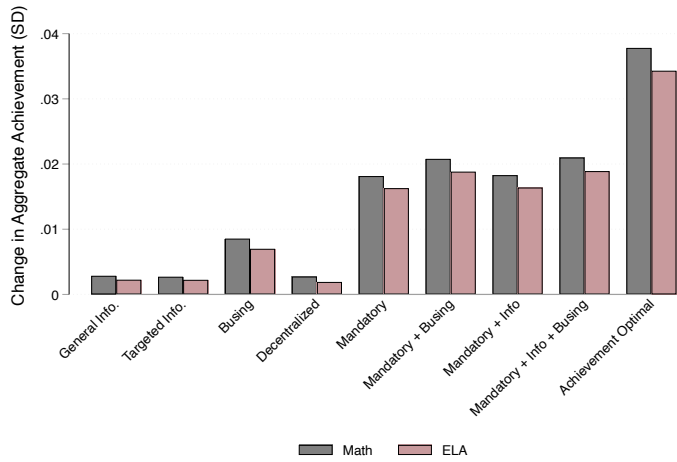
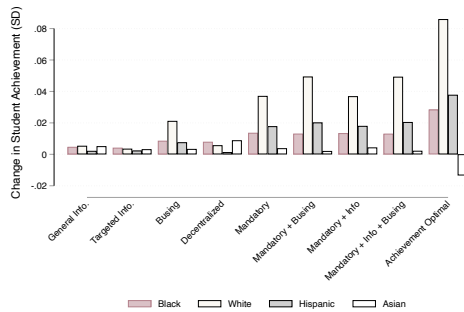


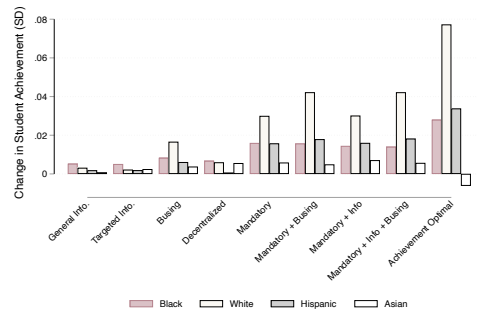
FIGURE A.13.—Choice School Peer Effect Analysis. *Note:* This figure reports lottery-based reduced form effects on own student achievement and school peer quality for Math and ELA in Panel (a) and Panel (b), respectively. Each marker in a given plot corresponds to a choice school, where the vertical axis reports the effect of winning an offer from the school on own achievement, and the horizontal axis reports the effect of winning an offer from the school on enrolled school peer quality. All estimated effects are conditional on lottery strata.



(a) Change in District-Level Average Achievement



(b) Change in Group-Specific Achievement (Math)



(c) Change in Group-Specific Achievement (ELA)

FIGURE A.14.—Estimated Effects of Counterfactual Policies on District-Level Achievement Overall and by Race. *Note:* This figure reports counterfactual achievement results relative to a baseline simulation of LAUSD’s current opt-in system. Panel (a) reports mean changes in district-level standardized achievement for Math and ELA, as identified in the figure legend. Panels (b) and (c) report group-specific changes in average achievement for Black, White, Hispanic, and Asian students in Math and ELA, respectively. We consider participation-targeted policies: General Information, which raises the taste parameter  $\theta_i$  for a randomly selected 50% of students; Targeted Information, which raises  $\theta_i$  for a randomly selected 50% of students enrolled in schools whose average achievement is below the specified cutoff of  $-0.30$  SD; and Busing, which sets distance costs to zero. We also consider two system-level designs: Decentralized choice, which allows families to submit multiple applications and hold simultaneous offers while schools run independent lotteries, and Mandatory application, which sets application costs to zero, requires a rank-ordered list (which may include the neighborhood school), and assigns seats via deferred acceptance. We also report combinations of these components (Mandatory + Busing; Mandatory + Information; Mandatory + Information + Busing) and an Achievement-Optimal benchmark that assigns students in order of modeled match quality. Each scenario is simulated 100 times, and we report average effects; for cases with school-run lotteries, we recompute equilibrium best-response admission probabilities under policy-induced demand (see Appendix Section C).

FIGURE A.15.—Changes in Spatial Demographic Shares Across Counterfactual Policies



*Notes:* Panels (a), (f), and (k) map baseline shares of low-income, Asian, and White students by attendance zone, respectively. Panels (b)–(e), (g)–(j), and (l)–(o) map counterfactual-specific changes in the corresponding demographic shares (counterfactual minus baseline). The counterfactual columns report General Information, Targeted Information, Mandatory, and Decentralized policies, respectively. Each polygon corresponds to an attendance zone boundary; shading indicates baseline levels in the first column and changes in the remaining columns. Within each demographic row, the four counterfactual panels use a common scale to facilitate comparisons across counterfactuals. The dark blue polygon in Panel (j) corresponds to a low-density region near the Hollywood Hills.

TABLE A.1  
SUMMARY STATISTICS FOR MAIN ANALYSIS SAMPLES

|                              | Sample                |                |                   |
|------------------------------|-----------------------|----------------|-------------------|
|                              | All<br>Grade 5<br>(1) | Lottery<br>(2) | Structural<br>(3) |
| Student Demographics         |                       |                |                   |
| Hispanic                     | 0.74                  | 0.57           | 0.76              |
| Black                        | 0.10                  | 0.14           | 0.09              |
| White                        | 0.09                  | 0.13           | 0.08              |
| Asian                        | 0.04                  | 0.12           | 0.04              |
| Female                       | 0.49                  | 0.51           | 0.50              |
| Poverty                      | 0.72                  | 0.63           | 0.70              |
| English Learner              | 0.27                  | 0.08           | 0.32              |
| Standardized Test Scores     |                       |                |                   |
| Baseline ELA                 | -0.01                 | 0.67           | -0.05             |
| Baseline Math                | -0.01                 | 0.65           | -0.04             |
| Distance to District Schools |                       |                |                   |
| Nearest                      | 0.75                  | 0.78           | 0.76              |
| Nearest Choice (Rel. Dist.)  | 0.57                  | 0.55           | 0.59              |
| Students                     | 593,550               | 27,203         | 337,661           |

*Note:* This table reports summary statistics for the various analysis samples we use throughout the paper. Column 1 reports statistics for the baseline sample consisting of all fifth-grade students in LAUSD during our sample period. Column 2 restricts the baseline sample to students who show up in our lottery sample, indicating they applied to a choice program and the choice program was oversubscribed. Column 3 restricts the baseline sample to students who meet the restrictions defined in Section 4 of the main paper.

TABLE A.2  
OVERSUBSCRIBED CHOICE PROGRAM LOTTERY BALANCE ANALYSIS

| Distance Quintile               | Q1                |              | Q2                |              | Q3                |              | Q4                |              | Q5                |              |
|---------------------------------|-------------------|--------------|-------------------|--------------|-------------------|--------------|-------------------|--------------|-------------------|--------------|
| Variable                        | Effect            | Control Mean | Effect            | Control Mean | Effect            | Control Mean | Effect            | Control Mean | Effect            | Control Mean |
| Female                          | -0.005<br>(0.014) | 0.507        | 0.021<br>(0.015)  | 0.511        | 0.004<br>(0.015)  | 0.506        | 0.004<br>(0.015)  | 0.497        | -0.006<br>(0.013) | 0.512        |
| Black                           | 0.005<br>(0.004)  | 0.057        | 0.002<br>(0.006)  | 0.065        | -0.010<br>(0.008) | 0.108        | -0.010<br>(0.009) | 0.168        | 0.010<br>(0.010)  | 0.277        |
| Hispanic                        | -0.006<br>(0.008) | 0.765        | -0.001<br>(0.009) | 0.648        | 0.009<br>(0.010)  | 0.545        | 0.011<br>(0.010)  | 0.475        | 0.000<br>(0.012)  | 0.501        |
| English Learner                 | 0.005<br>(0.007)  | 0.074        | -0.001<br>(0.007) | 0.071        | 0.011<br>(0.006)  | 0.054        | -0.003<br>(0.007) | 0.054        | -0.004<br>(0.006) | 0.061        |
| Special Education               | 0.000<br>(0.005)  | 0.027        | -0.003<br>(0.004) | 0.021        | -0.001<br>(0.004) | 0.020        | -0.002<br>(0.003) | 0.018        | 0.000<br>(0.003)  | 0.016        |
| Poverty                         | 0.018<br>(0.010)  | 0.758        | -0.019<br>(0.011) | 0.686        | 0.009<br>(0.012)  | 0.614        | -0.018<br>(0.012) | 0.573        | -0.003<br>(0.013) | 0.631        |
| Parent reports going to college | -0.001<br>(0.012) | 0.238        | -0.010<br>(0.012) | 0.254        | -0.004<br>(0.013) | 0.269        | 0.002<br>(0.013)  | 0.274        | -0.006<br>(0.012) | 0.262        |
| Speaks english at home          | 0.016<br>(0.012)  | 0.322        | -0.001<br>(0.012) | 0.356        | -0.007<br>(0.012) | 0.435        | -0.001<br>(0.013) | 0.508        | 0.017<br>(0.012)  | 0.519        |
| Speaks Spanish at home          | -0.021<br>(0.012) | 0.609        | 0.010<br>(0.011)  | 0.524        | 0.009<br>(0.010)  | 0.415        | -0.001<br>(0.011) | 0.353        | -0.011<br>(0.011) | 0.377        |
| Baseline Suspensions            | -0.002<br>(0.003) | 0.007        | 0.004<br>(0.002)  | 0.007        | 0.001<br>(0.004)  | 0.011        | 0.003<br>(0.004)  | 0.015        | -0.001<br>(0.005) | 0.024        |
| Baseline Math Score             | -0.013<br>(0.021) | 0.365        | -0.033<br>(0.022) | 0.480        | -0.008<br>(0.022) | 0.606        | -0.019<br>(0.025) | 0.569        | -0.061<br>(0.022) | 0.414        |
| Baseline ELA Score              | -0.031<br>(0.021) | 0.353        | -0.015<br>(0.022) | 0.482        | -0.001<br>(0.023) | 0.619        | -0.007<br>(0.023) | 0.605        | -0.030<br>(0.020) | 0.470        |
| Missing Baseline Score          | -0.003<br>(0.007) | 0.089        | -0.005<br>(0.006) | 0.099        | -0.002<br>(0.005) | 0.083        | -0.010<br>(0.005) | 0.094        | 0.000<br>(0.005)  | 0.088        |
| Joint test                      | 0.582             |              | 0.275             |              | 0.850             |              | 0.390             |              | 0.520             |              |
| Observations                    | 7,794             |              | 8,255             |              | 8,701             |              | 9,121             |              | 9,729             |              |

*Notes:* This table tests for lottery balance across baseline and other exogenous covariates. Columns in this table are grouped according to samples defined by quintiles of distance to choice school at baseline (Q1-Q5). Each row in this table denotes a predetermined covariate. Each cell under the columns labeled "Effect" gives the results of a separate regression of the relevant sub-sample. Standard errors, clustered at the level of the lottery, are included in parentheses. Columns labeled "Control Mean" give the mean of the variable in that row for that specific sample. The second to last row of this table gives  $p$ -values from a joint test that all of the coefficients in that column are identically zero. The final row gives the number of observations.

TABLE A.3  
ATTRITION RESULTS: IMPUTING TEST SCORES

|                               | <i>Quintile of Distance to Choice School</i> |                   |                   |                   |                   |
|-------------------------------|----------------------------------------------|-------------------|-------------------|-------------------|-------------------|
|                               | (1)                                          | (2)               | (3)               | (4)               | (5)               |
| Math                          | 0.039<br>(0.022)                             | -0.012<br>(0.021) | -0.027<br>(0.022) | -0.062<br>(0.020) | -0.058<br>(0.021) |
| Math (Missing Values Imputed) | 0.042<br>(0.020)                             | -0.003<br>(0.019) | -0.003<br>(0.019) | -0.047<br>(0.018) | -0.037<br>(0.019) |
| ELA                           | 0.003<br>(0.021)                             | -0.017<br>(0.020) | -0.039<br>(0.022) | -0.060<br>(0.020) | -0.056<br>(0.020) |
| ELA (Missing Values Imputed)  | 0.012<br>(0.019)                             | -0.004<br>(0.019) | -0.017<br>(0.019) | -0.045<br>(0.018) | -0.036<br>(0.018) |

*Notes:* This table explores sensitivity of the results in Figure 3 of the main paper to differential attrition. To do so, we use prior academic year test scores to impute values for the sample of students missing post-lottery math or ELA test scores. Specifically, we estimate univariate regressions of the form  $Y_{i,t} = \alpha + \beta Y_{i,t-1} + \epsilon_{i,t}$  where  $Y_{i,t}$  is the math or ELA score of student  $i$  in academic year  $t$  on the sample of students with non-missing test scores. We then impute missing test score outcomes for the students with missing test score outcomes using predicted values  $\hat{Y}_{i,t} = \hat{\alpha} + \hat{\beta} \hat{Y}_{i,t-1}$ . Students with non-missing test scores retain their original test score. Students missing both post-lottery and pre-lottery test scores ( $\approx 50\%$  of those missing a post-period score) are given the sample average ( $\hat{\alpha}$ ). We then re-estimate our baseline model using this constructed outcome. These results are contained in the rows labeled “Math (Missing Values Imputed)” and “ELA (Missing Values Imputed).” For comparison, the rows labeled “Math” and “ELA” give our baseline point estimates and standard errors from Figure 3 of the main paper. The point estimates are statistically indistinguishable across specifications.

TABLE A.4  
OVERSUBSCRIBED SCHOOL LOTTERY FIRST STAGE ESTIMATES BY QUINTILE

|               | <i>Quintile of Distance to Choice School</i> |                  |                  |                  |                  |
|---------------|----------------------------------------------|------------------|------------------|------------------|------------------|
|               | Q1                                           | Q2               | Q3               | Q4               | Q5               |
| Lottery Offer | 0.381<br>(0.022)                             | 0.409<br>(0.018) | 0.459<br>(0.015) | 0.434<br>(0.018) | 0.455<br>(0.019) |

*Note:* This table reports the first stage effects of receiving an offer. We report estimates for different quintiles of distance to choice school at baseline. The estimates come from a regression of enrolling in a choice school on lottery strata fixed effects, main effects for distance quintiles, and interactions of offers and distance quintiles. We report estimates of the interactions. Standard errors are clustered at the lottery level.

TABLE A.5  
LOTTERY ESTIMATES BY SUBGROUP

| Covariate         | Math                   |                        | ELA                    |                        |
|-------------------|------------------------|------------------------|------------------------|------------------------|
|                   | Focal                  | Complement             | Focal                  | Complement             |
|                   | Group<br>Covariate = 1 | Group<br>Covariate = 0 | Group<br>Covariate = 1 | Group<br>Covariate = 0 |
| High Achiever     | -0.041<br>(0.013)      | 0.025<br>(0.014)       | -0.049<br>(0.013)      | 0.016<br>(0.016)       |
| Female            | -0.014<br>(0.015)      | -0.038<br>(0.015)      | -0.031<br>(0.015)      | -0.039<br>(0.015)      |
| Poverty           | -0.016<br>(0.013)      | -0.064<br>(0.019)      | -0.030<br>(0.013)      | -0.059<br>(0.019)      |
| Special Education | 0.091<br>(0.079)       | -0.030<br>(0.011)      | 0.093<br>(0.074)       | -0.039<br>(0.011)      |
| Gifted            | -0.026<br>(0.025)      | -0.030<br>(0.012)      | -0.033<br>(0.027)      | -0.038<br>(0.012)      |
| English Learner   | 0.085<br>(0.031)       | -0.033<br>(0.011)      | 0.035<br>(0.032)       | -0.040<br>(0.011)      |
| English at home   | -0.065<br>(0.017)      | -0.000<br>(0.014)      | -0.074<br>(0.018)      | -0.009<br>(0.013)      |
| Spanish at home   | 0.013<br>(0.014)       | -0.061<br>(0.015)      | 0.005<br>(0.014)       | -0.070<br>(0.016)      |
| Born in USA       | -0.027<br>(0.011)      | -0.040<br>(0.040)      | -0.038<br>(0.011)      | -0.028<br>(0.038)      |

*Note:* This table presents heterogeneous effects by student subgroups. Each row in this table denotes one of the pre-determined characteristics on display in Figure A.4. With the exception of baseline achievement, all subgroups are based on binary pre-determined characteristics. In the case of achievement, we define high and low-ability sub-groups as those above and below the median of baseline achievement. We omit creating a subgroup based on suspensions, as it is a relatively rare outcome. All estimates are from separate subgroup specific models where we regress an achievement outcome on the lottery offer  $Z_i$  and lottery fixed effects. The first two columns correspond to treatment effects on Math; the second two columns correspond to treatment effects on ELA. Columns labeled “Focal Group” denote estimates from the sub-group of students where the covariate equals 1. “Complement Group” denotes estimates from the sub-group of students where the covariate equals zero. Standard errors, clustered at the level of the lottery, are included in parentheses. A visualization of these estimates, and their relationship to application rates by sub-group, is displayed in Appendix Figure A.6.

TABLE A.6  
COMPARISON OF ALTERNATIVE MODELS OF IDIOSYNCRATIC CHOICE SCHOOL PREFERENCES

|                         | One Type ( $K = 1$ ) | Two Types ( $K = 2$ ) | Three Types ( $K = 3$ ) |
|-------------------------|----------------------|-----------------------|-------------------------|
| Number of parameters    | 94                   | 97                    | 100                     |
| Log likelihood          | -64,260.9            | -62,872.4             | -62,625.2               |
| BIC                     | 129,587.9            | 126,844.9             | 126,384.6               |
| Likelihood ratio tests: |                      |                       |                         |
| $\chi^2$ statistic (df) |                      | 2,777.1 (3)           | 494.3 (3)               |
| $p$ -value              |                      | 0                     | 0                       |

*Note:* This table reports estimation statistics for different models we estimate. For each, we report the number of parameters associated with the model, the negative log-likelihood, a Bayesian Information Criterion (BIC) measure, and the statistic and  $p$ -value of a likelihood ratio test comparing a given model against the previous.

TABLE A.7  
DEMAND ESTIMATES: MIXTURE MODEL  
Panel (a): Estimates for Observable Parameters

|                                                  | Utility<br>(1)    | Distance Cost<br>(2) | Log Cost<br>(3)      |
|--------------------------------------------------|-------------------|----------------------|----------------------|
| Main Effects                                     | -0.897<br>[1.037] | 0.325<br>(0.005)     | 0.209<br>(0.014)     |
| Female                                           | 0.002<br>(0.023)  | 0.002<br>(0.003)     | 0.002<br>(0.003)     |
| Black                                            | -0.106<br>(0.057) | -0.016<br>(0.007)    | -0.035<br>(0.008)    |
| Hispanic                                         | -0.106<br>(0.039) | 0.012<br>(0.005)     | 0.065<br>(0.006)     |
| White                                            | -1.098<br>(0.045) | -0.131<br>(0.007)    | -0.083<br>(0.007)    |
| Poverty                                          | 0.070<br>(0.026)  | 0.015<br>(0.004)     | 0.014<br>(0.004)     |
| LEP                                              | -0.276<br>(0.040) | -0.016<br>(0.005)    | 0.010<br>(0.005)     |
| Speaks English at Home                           | -0.137<br>(0.031) | -0.024<br>(0.004)    | -0.010<br>(0.004)    |
| Baseline ELA                                     | 0.175<br>(0.018)  | 0.000<br>(0.002)     | -0.033<br>(0.003)    |
| Baseline Math                                    | 0.202<br>(0.018)  | 0.006<br>(0.002)     | -0.018<br>(0.003)    |
| Neighborhood Median Income                       | -0.059<br>(0.005) | 0.001<br>(0.001)     | 0.005<br>(0.001)     |
| Baseline Choice Enrollment                       | 1.275<br>(0.038)  | -0.076<br>(0.004)    | -0.097<br>(0.007)    |
| Baseline Peer Quality                            | 0.123<br>(0.028)  | -0.026<br>(0.004)    | -0.026<br>(0.005)    |
| Panel (b): Estimates for Unobservable Parameters |                   |                      |                      |
|                                                  | $\mu$<br>(1)      | $\sigma$<br>(2)      | $Pr(K_i = k)$<br>(3) |
| Type 1                                           | -0.876<br>(0.046) | 0.189<br>(0.062)     | 0.446                |
| Type 2                                           | 0.430<br>(0.042)  | 0.125<br>(0.043)     | 0.488                |
| Type 3                                           | 2.749<br>(1.051)  | 0.541<br>(0.039)     | 0.066                |
| Cost Heterogeneity                               |                   | 0.363<br>(0.006)     |                      |

*Note:* This table reports demand model results (Section 7.2 of the main paper) estimated via simulated maximum likelihood with 300 draws for taste heterogeneity ( $\theta_i$ ) and cost heterogeneity ( $\eta_i$ ). Panel (a) reports observable heterogeneity: the first row gives main effects for distance and log cost; school mean utilities are shown as averages with noise-adjusted SDs in brackets; remaining rows are heterogeneity terms. Panel (b) reports unobservables: a mixture-of-normals for tastes (means in col. 1, SDs in col. 2, type probabilities in col. 3) and mean-zero normal costs with the SD in col. 2. Standard errors are in parentheses.

TABLE A.8  
STUDENT ACHIEVEMENT MODEL ESTIMATES (TYPE PROBABILITIES SPECIFICATION)

|                                                              | Model: Type Probabilities |                   |                   |                   |
|--------------------------------------------------------------|---------------------------|-------------------|-------------------|-------------------|
|                                                              | Math                      |                   | ELA               |                   |
|                                                              | $\gamma_0$<br>(1)         | $\gamma_c$<br>(2) | $\gamma_0$<br>(3) | $\gamma_c$<br>(4) |
| <i>Panel A. Average choice effects</i>                       |                           |                   |                   |                   |
| Main effects                                                 |                           | 0.290 [0.160]     |                   | 0.243 [0.113]     |
| <i>Panel B. Demographic and socioeconomic controls</i>       |                           |                   |                   |                   |
| Female                                                       | -0.022 (0.002)            | -0.009 (0.007)    | 0.111 (0.002)     | 0.002 (0.006)     |
| Black                                                        | -0.196 (0.011)            | 0.029 (0.022)     | -0.161 (0.010)    | 0.027 (0.020)     |
| White                                                        | -0.004 (0.012)            | -0.027 (0.021)    | 0.001 (0.009)     | -0.027 (0.021)    |
| Hispanic                                                     | -0.149 (0.010)            | -0.004 (0.020)    | -0.119 (0.008)    | 0.012 (0.019)     |
| Asian                                                        | 0.211 (0.014)             | -0.062 (0.025)    | 0.075 (0.010)     | -0.042 (0.022)    |
| Poverty                                                      | 0.010 (0.003)             | -0.011 (0.010)    | 0.003 (0.002)     | -0.006 (0.007)    |
| LEP                                                          | -0.087 (0.004)            | 0.037 (0.013)     | -0.171 (0.003)    | 0.010 (0.014)     |
| Median income                                                |                           | -0.005 (0.002)    |                   | -0.006 (0.002)    |
| English home                                                 | -0.070 (0.004)            | 0.009 (0.012)     | -0.059 (0.004)    | 0.044 (0.010)     |
| <i>Panel C. Baseline achievement and peer quality</i>        |                           |                   |                   |                   |
| Baseline Math                                                | 0.526 (0.004)             | -0.038 (0.009)    | 0.182 (0.002)     | -0.023 (0.006)    |
| Baseline ELA                                                 | 0.210 (0.003)             | 0.012 (0.007)     | 0.591 (0.003)     | -0.016 (0.008)    |
| Baseline peer quality                                        | -0.149 (0.012)            | 0.032 (0.018)     | -0.095 (0.009)    | 0.024 (0.014)     |
| <i>Panel D. Type-probability control-function parameters</i> |                           |                   |                   |                   |
| Pr(Type 2)                                                   | -0.053 (0.032)            | -0.167 (0.047)    | -0.006 (0.023)    | -0.118 (0.040)    |
| Pr(Type 3)                                                   | 0.076 (0.015)             | -0.385 (0.039)    | 0.102 (0.012)     | -0.309 (0.032)    |

*Note:* Estimates from an alternative type-specific specification of the achievement model. Let  $T_{ik} = \mathbf{1}\{\tau_i = k\}$  denote membership in latent preference type  $k$ , where the three types correspond to the mixture components used to approximate the distribution of  $\theta_i$  in the demand model. The specification replaces  $g_j(\theta_i)$  in the main paper achievement model with  $\sum_{k \neq 1} \gamma_k T_{ik} + \sum_{k \neq 1} \gamma_{ck} T_{ik} \times \mathbf{1}\{j > 0\}$ , using Type 1 as the omitted category. Thus,  $\gamma_k$  captures type-specific differences in achievement levels and  $\gamma_{ck}$  captures type-specific differences in choice-school treatment effects relative to Type 1. Odd columns report baseline coefficients ( $\gamma_0$ ); even columns report choice-enrollee interactions ( $\gamma_c$ ). “Main effects” are enrollment-weighted average choice effects for Type 1; noise-adjusted standard deviations are in brackets. All specifications include census-block, year, and subdistrict fixed effects. Robust standard errors clustered by enrolled school are in parentheses.

TABLE A.9  
OUTCOME MODEL BOOTSTRAPPED ESTIMATES

|                                     | Model 1: Type Probabilities              |                                         |                                          |                                         | Model 2: Linear Control Function ( $\theta_i$ ) |                                         |                                          |                                         |
|-------------------------------------|------------------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------|-------------------------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------|
|                                     | Math                                     |                                         | ELA                                      |                                         | Math                                            |                                         | ELA                                      |                                         |
|                                     | Neighborhood<br>School<br>( $\gamma_0$ ) | Choice<br>School<br>Diff ( $\gamma_c$ ) | Neighborhood<br>School<br>( $\gamma_0$ ) | Choice<br>School<br>Diff ( $\gamma_c$ ) | Neighborhood<br>School<br>( $\gamma_0$ )        | Choice<br>School<br>Diff ( $\gamma_c$ ) | Neighborhood<br>School<br>( $\gamma_0$ ) | Choice<br>School<br>Diff ( $\gamma_c$ ) |
|                                     | (1)                                      | (2)                                     | (3)                                      | (4)                                     | (5)                                             | (6)                                     | (7)                                      | (8)                                     |
| Main Effects                        |                                          | 0.252<br>(0.003)<br>[0.173]             |                                          | 0.212<br>(0.001)<br>[0.117]             |                                                 | 0.223<br>(0.002)<br>[0.171]             |                                          | 0.198<br>(0.001)<br>[0.115]             |
| Female                              | -0.024<br>(0.002)                        | -0.012<br>(0.007)                       | 0.109<br>(0.002)                         | 0.001<br>(0.006)                        | -0.024<br>(0.002)                               | -0.012<br>(0.007)                       | 0.109<br>(0.002)                         | 0.001<br>(0.006)                        |
| Black                               | -0.276<br>(0.016)                        | 0.080<br>(0.026)                        | -0.222<br>(0.013)                        | 0.058<br>(0.020)                        | -0.276<br>(0.016)                               | 0.080<br>(0.026)                        | -0.222<br>(0.013)                        | 0.058<br>(0.020)                        |
| White                               | 0.092<br>(0.018)                         | -0.072<br>(0.027)                       | 0.064<br>(0.013)                         | -0.052<br>(0.022)                       | 0.092<br>(0.018)                                | -0.076<br>(0.027)                       | 0.064<br>(0.013)                         | -0.054<br>(0.022)                       |
| Hispanic                            | -0.174<br>(0.013)                        | 0.030<br>(0.022)                        | -0.142<br>(0.010)                        | 0.033<br>(0.016)                        | -0.175<br>(0.013)                               | 0.026<br>(0.022)                        | -0.142<br>(0.010)                        | 0.031<br>(0.016)                        |
| Asian                               | 0.270<br>(0.017)                         | -0.065<br>(0.025)                       | 0.107<br>(0.011)                         | -0.052<br>(0.020)                       | 0.270<br>(0.017)                                | -0.063<br>(0.026)                       | 0.106<br>(0.011)                         | -0.050<br>(0.020)                       |
| Poverty                             | -0.016<br>(0.004)                        | 0.000<br>(0.009)                        | -0.021<br>(0.003)                        | 0.001<br>(0.007)                        | -0.016<br>(0.004)                               | -0.001<br>(0.009)                       | -0.021<br>(0.003)                        | 0.000<br>(0.007)                        |
| LEP                                 | -0.084<br>(0.004)                        | 0.033<br>(0.013)                        | -0.170<br>(0.004)                        | 0.001<br>(0.014)                        | -0.083<br>(0.004)                               | 0.037<br>(0.013)                        | -0.170<br>(0.004)                        | 0.003<br>(0.014)                        |
| Median Income                       |                                          | 0.015<br>(0.002)                        |                                          | 0.013<br>(0.002)                        |                                                 | 0.015<br>(0.002)                        |                                          | 0.012<br>(0.002)                        |
| English Home                        | -0.037<br>(0.005)                        | -0.020<br>(0.012)                       | -0.028<br>(0.005)                        | 0.020<br>(0.011)                        | -0.037<br>(0.005)                               | -0.020<br>(0.012)                       | -0.027<br>(0.005)                        | 0.020<br>(0.011)                        |
| Baseline Math                       | 0.518<br>(0.004)                         | -0.032<br>(0.009)                       | 0.173<br>(0.003)                         | -0.016<br>(0.006)                       | 0.518<br>(0.004)                                | -0.031<br>(0.010)                       | 0.173<br>(0.003)                         | -0.016<br>(0.006)                       |
| Baseline ELA                        | 0.227<br>(0.003)                         | 0.004<br>(0.006)                        | 0.607<br>(0.003)                         | -0.024<br>(0.008)                       | 0.227<br>(0.003)                                | 0.006<br>(0.006)                        | 0.607<br>(0.003)                         | -0.024<br>(0.008)                       |
| Baseline Peer Quality               | -0.063<br>(0.013)                        | -0.035<br>(0.021)                       | -0.029<br>(0.010)                        | -0.025<br>(0.017)                       | -0.062<br>(0.013)                               | -0.032<br>(0.022)                       | -0.028<br>(0.010)                        | -0.024<br>(0.017)                       |
| Pr(Type 2)                          | -0.026<br>(0.042)                        | -0.152<br>(0.055)                       | 0.001<br>(0.030)                         | -0.108<br>(0.042)                       |                                                 |                                         |                                          |                                         |
| Pr(Type 3)                          | 0.088<br>(0.022)                         | -0.341<br>(0.045)                       | 0.105<br>(0.015)                         | -0.269<br>(0.032)                       |                                                 |                                         |                                          |                                         |
| Choice School Preference $\theta_i$ |                                          |                                         |                                          |                                         | 0.025<br>(0.005)                                | -0.068<br>(0.010)                       | 0.029<br>(0.004)                         | -0.060<br>(0.007)                       |
| Neighborhood Effects                | ✓                                        |                                         | ✓                                        |                                         | ✓                                               |                                         | ✓                                        |                                         |
| Year Effects                        | ✓                                        |                                         | ✓                                        |                                         | ✓                                               |                                         | ✓                                        |                                         |
| Sub-district Effects                | ✓                                        |                                         | ✓                                        |                                         | ✓                                               |                                         | ✓                                        |                                         |
| Bootstrap Iterations                | 100                                      |                                         | 100                                      |                                         | ✓                                               |                                         | ✓                                        |                                         |

*Note:* This table reports bootstrapped estimates of the achievement model in Section 7.3 of the main paper for Model 1 (Cols. 1–4) and Model 2 (Cols. 5–8). This analysis addresses concern over estimation error in either type probability estimates (in Model 1) or the posterior  $\theta_i$  (in Model 2) estimates. We assume a three-type ( $K = 3$ ) mixture model for the distribution of  $\theta_i$ . For each bootstrap iteration, we draw from the asymptotic distribution of the underlying demand estimates, recompute either the posterior type probabilities (Model 1) or posterior means of  $\theta_i$  (Model 2), and then re-estimate the outcome model. The entries in the table are bootstrap averages of the corresponding point estimates; bootstrapped standard errors clustered by enrolled school are reported in parentheses. Cols. 1, 3, 5, and 7 report baseline coefficients ( $\gamma_0$ ) for the covariates listed in each row; Cols. 2, 4, 6, and 8 report the corresponding choice-enrollee interactions ( $\gamma_c$ ). The “Main effects” row gives the enrollment-weighted average effect of attending a choice school; noise-adjusted SDs across schools are shown in brackets. All specifications include Census-block, year, and sub-district fixed effects.

TABLE A.10  
OUTCOME MODEL ESTIMATES WITH NEIGHBORHOOD SCHOOL VAM HETEROGENEITY

|                                     | Model 1: Type Probabilities            |                                       |                                        |                                       | Model 2: Linear Control Function       |                                       |                                        |                                       |
|-------------------------------------|----------------------------------------|---------------------------------------|----------------------------------------|---------------------------------------|----------------------------------------|---------------------------------------|----------------------------------------|---------------------------------------|
|                                     | Math                                   |                                       | ELA                                    |                                       | Math                                   |                                       | ELA                                    |                                       |
|                                     | Neighborhood School ( $\gamma_0$ ) (1) | Choice School Diff ( $\gamma_c$ ) (2) | Neighborhood School ( $\gamma_0$ ) (3) | Choice School Diff ( $\gamma_c$ ) (4) | Neighborhood School ( $\gamma_0$ ) (5) | Choice School Diff ( $\gamma_c$ ) (6) | Neighborhood School ( $\gamma_0$ ) (7) | Choice School Diff ( $\gamma_c$ ) (8) |
| Main Effects                        |                                        | 0.293<br>[0.160]                      |                                        | 0.243<br>[0.114]                      |                                        | 0.259<br>[0.159]                      |                                        | 0.229<br>[0.112]                      |
| Female                              | -0.022<br>(0.002)                      | -0.009<br>(0.007)                     | 0.111<br>(0.002)                       | 0.001<br>(0.006)                      | -0.022<br>(0.002)                      | -0.010<br>(0.007)                     | 0.111<br>(0.002)                       | 0.001<br>(0.006)                      |
| Black                               | -0.196<br>(0.011)                      | 0.027<br>(0.022)                      | -0.161<br>(0.010)                      | 0.025<br>(0.020)                      | -0.197<br>(0.011)                      | 0.026<br>(0.022)                      | -0.161<br>(0.010)                      | 0.025<br>(0.020)                      |
| White                               | -0.005<br>(0.012)                      | -0.025<br>(0.021)                     | 0.001<br>(0.009)                       | -0.025<br>(0.021)                     | -0.004<br>(0.011)                      | -0.030<br>(0.021)                     | 0.001<br>(0.009)                       | -0.027<br>(0.021)                     |
| Hispanic                            | -0.149<br>(0.010)                      | -0.002<br>(0.020)                     | -0.119<br>(0.008)                      | 0.013<br>(0.018)                      | -0.150<br>(0.010)                      | -0.006<br>(0.020)                     | -0.119<br>(0.008)                      | 0.011<br>(0.018)                      |
| Asian                               | 0.211<br>(0.014)                       | -0.060<br>(0.025)                     | 0.075<br>(0.009)                       | -0.041<br>(0.022)                     | 0.210<br>(0.014)                       | -0.057<br>(0.025)                     | 0.074<br>(0.009)                       | -0.039<br>(0.022)                     |
| Poverty                             | 0.010<br>(0.003)                       | -0.011<br>(0.009)                     | 0.003<br>(0.002)                       | -0.007<br>(0.007)                     | 0.010<br>(0.003)                       | -0.013<br>(0.010)                     | 0.003<br>(0.002)                       | -0.007<br>(0.007)                     |
| LEP                                 | -0.087<br>(0.004)                      | 0.036<br>(0.013)                      | -0.171<br>(0.003)                      | 0.008<br>(0.014)                      | -0.087<br>(0.004)                      | 0.041<br>(0.013)                      | -0.171<br>(0.003)                      | 0.012<br>(0.014)                      |
| Median Income                       |                                        | -0.004<br>(0.002)                     |                                        | -0.004<br>(0.002)                     |                                        | -0.005<br>(0.002)                     |                                        | -0.005<br>(0.002)                     |
| English Home                        | -0.071<br>(0.004)                      | 0.010<br>(0.012)                      | -0.059<br>(0.004)                      | 0.045<br>(0.010)                      | -0.070<br>(0.004)                      | 0.009<br>(0.012)                      | -0.059<br>(0.004)                      | 0.045<br>(0.010)                      |
| Baseline Math                       | 0.526<br>(0.004)                       | -0.037<br>(0.009)                     | 0.182<br>(0.002)                       | -0.023<br>(0.006)                     | 0.526<br>(0.004)                       | -0.036<br>(0.009)                     | 0.182<br>(0.002)                       | -0.023<br>(0.006)                     |
| Baseline ELA                        | 0.210<br>(0.003)                       | 0.011<br>(0.007)                      | 0.591<br>(0.003)                       | -0.016<br>(0.008)                     | 0.210<br>(0.003)                       | 0.013<br>(0.007)                      | 0.591<br>(0.003)                       | -0.016<br>(0.008)                     |
| Baseline Peer Quality               | -0.150<br>(0.012)                      | 0.030<br>(0.018)                      | -0.096<br>(0.009)                      | 0.024<br>(0.014)                      | -0.148<br>(0.012)                      | 0.033<br>(0.019)                      | -0.095<br>(0.009)                      | 0.024<br>(0.014)                      |
| Neighborhood School VAM             | 0.056<br>(0.018)                       | -0.049<br>(0.044)                     | 0.071<br>(0.017)                       | -0.112<br>(0.044)                     | 0.056<br>(0.018)                       | -0.046<br>(0.044)                     | 0.071<br>(0.017)                       | -0.114<br>(0.044)                     |
| Pr(Type 2)                          | -0.073<br>(0.034)                      | -0.134<br>(0.046)                     | -0.020<br>(0.025)                      | -0.100<br>(0.041)                     |                                        |                                       |                                        |                                       |
| Pr(Type 3)                          | 0.064<br>(0.017)                       | -0.382<br>(0.040)                     | 0.094<br>(0.013)                       | -0.303<br>(0.032)                     |                                        |                                       |                                        |                                       |
| Choice School Preference $\theta_i$ |                                        |                                       |                                        |                                       | 0.022<br>(0.003)                       | -0.077<br>(0.009)                     | 0.028<br>(0.003)                       | -0.069<br>(0.006)                     |
| Neighborhood Effects                | ✓                                      |                                       | ✓                                      |                                       | ✓                                      |                                       | ✓                                      |                                       |
| Year Effects                        | ✓                                      |                                       | ✓                                      |                                       | ✓                                      |                                       | ✓                                      |                                       |
| Sub-district Effects                | ✓                                      |                                       | ✓                                      |                                       | ✓                                      |                                       | ✓                                      |                                       |

*Note:* This table reports estimates from the type-probability specification (Cols. 1–4) and the linear control function specification (Cols. 5–8). The control functions—both posterior type probabilities and posterior  $\theta$ —are based on an auxiliary demand model that adds time-varying school observables to the utility specification including neighborhood school quality. The odd columns (1, 3, 5, and 7) report baseline coefficients ( $\gamma_0$ ); the even columns (2, 4, 6, and 8) report choice-enrollee interactions ( $\gamma_c$ ). The total effect for choice students is  $\gamma_0 + \gamma_c$ . The “Main effects” row gives the enrollment-weighted average choice effect; noise-adjusted SDs are in brackets. All specs include Census-block, year, and sub-district fixed effects. No main effect for Median Income is shown because it is collinear with block FEs; its interaction is identified from within-block variation in the choice indicator.

TABLE A.11  
OUTCOME MODEL ESTIMATES WITH DISTANCE HETEROGENEITY

|                         | Model 1: Type Probabilities |                     |                     |                     | Model 2: Linear Control Function |                     |                     |                     |
|-------------------------|-----------------------------|---------------------|---------------------|---------------------|----------------------------------|---------------------|---------------------|---------------------|
|                         | Math                        |                     | ELA                 |                     | Math                             |                     | ELA                 |                     |
|                         | Neighborhood School         | Choice School       | Neighborhood School | Choice School       | Neighborhood School              | Choice School       | Neighborhood School | Choice School       |
|                         | ( $\gamma_0$ )              | Diff ( $\gamma_c$ ) | ( $\gamma_0$ )      | Diff ( $\gamma_c$ ) | ( $\gamma_0$ )                   | Diff ( $\gamma_c$ ) | ( $\gamma_0$ )      | Diff ( $\gamma_c$ ) |
|                         | (1)                         | (2)                 | (3)                 | (4)                 | (5)                              | (6)                 | (7)                 | (8)                 |
| Main Effects            |                             | 0.292<br>[0.160]    |                     | 0.245<br>[0.113]    |                                  | 0.258<br>[0.158]    |                     | 0.231<br>[0.112]    |
| Female                  | -0.022<br>(0.002)           | -0.009<br>(0.007)   | 0.111<br>(0.002)    | 0.001<br>(0.006)    | -0.022<br>(0.002)                | -0.010<br>(0.007)   | 0.111<br>(0.002)    | 0.001<br>(0.006)    |
| Black                   | -0.196<br>(0.011)           | 0.028<br>(0.022)    | -0.161<br>(0.010)   | 0.025<br>(0.020)    | -0.197<br>(0.011)                | 0.027<br>(0.022)    | -0.161<br>(0.010)   | 0.024<br>(0.020)    |
| White                   | -0.005<br>(0.012)           | -0.026<br>(0.021)   | 0.001<br>(0.009)    | -0.025<br>(0.021)   | -0.004<br>(0.011)                | -0.030<br>(0.021)   | 0.001<br>(0.009)    | -0.027<br>(0.021)   |
| Hispanic                | -0.149<br>(0.010)           | -0.001<br>(0.020)   | -0.119<br>(0.008)   | 0.013<br>(0.019)    | -0.150<br>(0.010)                | -0.006<br>(0.020)   | -0.119<br>(0.008)   | 0.011<br>(0.019)    |
| Asian                   | 0.211<br>(0.014)            | -0.060<br>(0.025)   | 0.075<br>(0.009)    | -0.040<br>(0.022)   | 0.210<br>(0.014)                 | -0.057<br>(0.025)   | 0.074<br>(0.009)    | -0.038<br>(0.022)   |
| Poverty                 | 0.010<br>(0.003)            | -0.011<br>(0.009)   | 0.003<br>(0.002)    | -0.007<br>(0.007)   | 0.010<br>(0.003)                 | -0.013<br>(0.010)   | 0.003<br>(0.002)    | -0.007<br>(0.007)   |
| LEP                     | -0.087<br>(0.004)           | 0.036<br>(0.013)    | -0.171<br>(0.003)   | 0.008<br>(0.014)    | -0.087<br>(0.004)                | 0.041<br>(0.013)    | -0.171<br>(0.003)   | 0.012<br>(0.014)    |
| Median Income           |                             | -0.004<br>(0.002)   |                     | -0.004<br>(0.002)   |                                  | -0.005<br>(0.002)   |                     | -0.005<br>(0.002)   |
| English Home            | -0.071<br>(0.004)           | 0.010<br>(0.012)    | -0.059<br>(0.004)   | 0.045<br>(0.010)    | -0.070<br>(0.004)                | 0.009<br>(0.012)    | -0.059<br>(0.004)   | 0.045<br>(0.010)    |
| Baseline Math           | 0.526<br>(0.004)            | -0.038<br>(0.009)   | 0.182<br>(0.002)    | -0.023<br>(0.006)   | 0.526<br>(0.004)                 | -0.037<br>(0.009)   | 0.182<br>(0.002)    | -0.023<br>(0.006)   |
| Baseline ELA            | 0.210<br>(0.003)            | 0.011<br>(0.007)    | 0.591<br>(0.003)    | -0.016<br>(0.008)   | 0.210<br>(0.003)                 | 0.013<br>(0.007)    | 0.591<br>(0.003)    | -0.016<br>(0.008)   |
| Baseline Peer Quality   | -0.150<br>(0.012)           | 0.030<br>(0.018)    | -0.096<br>(0.009)   | 0.024<br>(0.014)    | -0.148<br>(0.012)                | 0.033<br>(0.019)    | -0.095<br>(0.009)   | 0.024<br>(0.014)    |
| Neighborhood School VAM | 0.056<br>(0.018)            | -0.072<br>(0.033)   | 0.071<br>(0.017)    | -0.119<br>(0.031)   | 0.056<br>(0.018)                 | -0.070<br>(0.033)   | 0.071<br>(0.017)    | -0.120<br>(0.031)   |
| Relative Distance       |                             | 0.004<br>(0.006)    |                     | -0.004<br>(0.005)   |                                  | 0.004<br>(0.006)    |                     | -0.004<br>(0.005)   |
| Pr(Type 2)              | -0.074<br>(0.034)           | -0.135<br>(0.046)   | -0.020<br>(0.025)   | -0.100<br>(0.041)   |                                  |                     |                     |                     |
| Pr(Type 3)              | 0.064<br>(0.017)            | -0.384<br>(0.040)   | 0.094<br>(0.013)    | -0.301<br>(0.033)   |                                  |                     |                     |                     |
| $\theta_i$              |                             |                     |                     |                     | 0.022<br>(0.003)                 | -0.077<br>(0.009)   | 0.028<br>(0.003)    | -0.069<br>(0.006)   |
| Neighborhood Effects    | ✓                           |                     | ✓                   |                     | ✓                                |                     | ✓                   |                     |
| Year Effects            | ✓                           |                     | ✓                   |                     | ✓                                |                     | ✓                   |                     |
| Sub-district Effects    | ✓                           |                     | ✓                   |                     | ✓                                |                     | ✓                   |                     |

*Note:* This table reports estimates of the type-probability specification (Cols. 1–4) and the linear control specification (Cols. 5–8), augmented to allow treatment-effect heterogeneity in relative distance to the nearest choice school. The control functions—both posterior type probabilities and posterior  $\theta$ —are based on an auxiliary demand model that adds time-varying school observables to the utility specification and a shifter for busing eligibility that is excluded from the outcome equation. Therefore, identification of the selection coefficients comes from variation in lottery offers and shifts in access to transportation. Cols. 1, 3, 5, and 7 report baseline coefficients ( $\gamma_0$ ) for the covariates listed in each row; Cols. 2, 4, 6, and 8 report the corresponding choice-enrollee interactions ( $\gamma_c$ ). The “Main effects” row gives the enrollment-weighted average effect of attending a choice school; noise-adjusted SDs are in brackets. All specifications include Census-block, year, and sub-district fixed effects. Robust SEs clustered by enrolled school are in parentheses.

TABLE A.12  
ANALYSIS OF PARTICIPATION REFORMS AND ALTERNATIVE CHOICE SYSTEMS - AUXILIARY MODEL I

Panel (a): Baseline Simulation

|                    | Participation (%) |                         | All Student Achievement ( $\sigma$ ) |         | Choice Enrollee TOT ( $\sigma$ ) |         | Demographic Group Choice Enrollment (%) |           |              |            |              |
|--------------------|-------------------|-------------------------|--------------------------------------|---------|----------------------------------|---------|-----------------------------------------|-----------|--------------|------------|--------------|
|                    | Apply (1)         | Choice Seats Filled (2) | Math (3)                             | ELA (4) | Math (5)                         | ELA (6) | Asian (7)                               | Black (8) | Hispanic (9) | White (10) | Low SES (11) |
| Opt-in (Voluntary) | 15.30             | 54.28                   | .                                    | .       | 0.122                            | 0.093   | 17.04                                   | 7.92      | 4.77         | 13.66      | 5.55         |

Panel (b): Alternative Policies/System Impacts (Relative to Baseline)

|                                      | Participation Gains (%) |                         | All Student Achievement Gains ( $\sigma$ ) |         | Choice Enrollee TOT Gains ( $\sigma$ ) |         | Choice Enrollee Demographic Gains (%) |           |              |            |              |
|--------------------------------------|-------------------------|-------------------------|--------------------------------------------|---------|----------------------------------------|---------|---------------------------------------|-----------|--------------|------------|--------------|
|                                      | Apply (1)               | Choice Seats Filled (2) | Math (3)                                   | ELA (4) | Math (5)                               | ELA (6) | Asian (7)                             | Black (8) | Hispanic (9) | White (10) | Low SES (11) |
| Broad Information                    | + 3.76                  | + 20.42                 | + 0.001                                    | + 0.002 | + 0.014                                | + 0.013 | + 4.78                                | + 3.21    | + 1.95       | + 4.60     | + 2.20       |
| Targeted Information                 | + 3.81                  | + 21.33                 | + 0.001                                    | + 0.002 | + 0.012                                | + 0.012 | + 4.09                                | + 3.50    | + 2.15       | + 4.21     | + 2.38       |
| Busing Program                       | + 16.71                 | + 37.96                 | + 0.007                                    | + 0.006 | + 0.039                                | + 0.036 | + 3.99                                | + 4.09    | + 3.92       | + 11.16    | + 4.68       |
| Decentralized                        | -4.85                   | + 27.49                 | + 0.001                                    | + 0.000 | + 0.005                                | + 0.005 | + 9.38                                | + 5.31    | + 2.25       | + 6.97     | + 2.82       |
| Mandatory App.                       | + 65.17                 | + 43.26                 | + 0.017                                    | + 0.015 | + 0.112                                | + 0.103 | -13.98                                | + 3.02    | + 4.91       | + 15.79    | + 5.27       |
| Mandatory App. + Busing              | + 82.94                 | + 45.44                 | + 0.021                                    | + 0.019 | + 0.135                                | + 0.124 | -14.58                                | + 2.00    | + 5.25       | + 18.82    | + 5.64       |
| Mandatory App. + Broad Info          | + 65.90                 | + 43.51                 | + 0.018                                    | + 0.015 | + 0.113                                | + 0.103 | -14.02                                | + 3.16    | + 4.92       | + 16.05    | + 5.31       |
| Mandatory App. + Broad Info + Busing | + 83.00                 | + 45.45                 | + 0.020                                    | + 0.018 | + 0.135                                | + 0.124 | -14.60                                | + 1.97    | + 5.25       | + 18.84    | + 5.65       |
| Maximized Achievement                | —                       | + 45.72                 | + 0.037                                    | + 0.031 | + 0.267                                | + 0.240 | -16.66                                | + 11.67   | + 4.95       | + 19.24    | + 4.94       |

*Note:* This table reports simulated results from a model of application, enrollment, and achievement, with modifications to the achievement model demand models relative to our preferred model in the main body of the paper. In particular, on the demand side, we enrich utility by allowing for time-varying school observables and student-specific neighborhood quality. For the outcome model, we allow for treatment effect heterogeneity with respect to neighborhood school quality. The corresponding outcome model estimates are reported in Appendix Table A.10. Panel (a) reports results for the baseline opt-in system. Panel (b) reports counterfactual impacts, expressed as the difference relative to the baseline scenario. We report the percent of students that apply to choice schools as their most-preferred option (Column 1), and the percent of choice school seats that are filled (Column 2). For achievement, we report district-wide achievement (Columns 3 and 4) and treatment on the treated estimates (Columns 5 and 6). Demographic outcomes include the percent of students within each racial or income-group enrolling in a choice school (Columns 7-11). The General Information scenario increases the  $\theta_i$  of a randomly selected 50% of students in the district, while the Targeted Information scenario increases the  $\theta_i$  of a randomly selected 50% of students who are enrolled in schools whose average achievement is below median. The No Travel Costs scenario mirrors a generous busing policy that sets travel costs to zero. The Decentralized scenario allows families to submit multiple independent applications to schools, which then run their own lotteries, potentially generating multiple simultaneous offers. The Mandatory scenario eliminates application costs, allowing families to rank their neighborhood school as their most-preferred option and assignments are done via a deferred acceptance mechanism. For each counterfactual, we simulate the economy 100 times. For scenarios that retain school-run lotteries (information, busing, and decentralized choice), we calculate equilibrium best response admissions probabilities that schools would set under the changed demand implied by the policy (see Appendix Section C for details).

TABLE A.13  
ANALYSIS OF PARTICIPATION REFORMS AND ALTERNATIVE CHOICE SYSTEMS - AUXILIARY MODEL II

Panel (a): Baseline Simulation

|                    | Participation (%) |                         | All Student Achievement ( $\sigma$ ) |         | Choice Enrollee TOT ( $\sigma$ ) |         | Choice Enrollee Demographics (%) |           |              |            |              |
|--------------------|-------------------|-------------------------|--------------------------------------|---------|----------------------------------|---------|----------------------------------|-----------|--------------|------------|--------------|
|                    | Apply (1)         | Choice Seats Filled (2) | Math (3)                             | ELA (4) | Math (5)                         | ELA (6) | Asian (7)                        | Black (8) | Hispanic (9) | White (10) | Low SES (11) |
| Opt-in (Voluntary) | 15.29             | 54.17                   | .                                    | .       | 0.123                            | 0.093   | 17.43                            | 7.98      | 4.74         | 13.54      | 5.56         |

Panel (b): Alternative Policies/System Impacts (Relative to Baseline)

|                                      | Participation Gains (%) |                         | All Student Achievement Gains ( $\sigma$ ) |         | Choice Enrollee TOT Gains ( $\sigma$ ) |         | Demographic Group Choice Enrollment (%) |           |              |            |              |
|--------------------------------------|-------------------------|-------------------------|--------------------------------------------|---------|----------------------------------------|---------|-----------------------------------------|-----------|--------------|------------|--------------|
|                                      | Apply (1)               | Choice Seats Filled (2) | Math (3)                                   | ELA (4) | Math (5)                               | ELA (6) | Asian (7)                               | Black (8) | Hispanic (9) | White (10) | Low SES (11) |
| Broad Information                    | + 3.65                  | + 20.19                 | + 0.002                                    | + 0.004 | + 0.014                                | + 0.013 | + 4.36                                  | + 3.26    | + 1.91       | + 4.70     | + 2.14       |
| Targeted Information                 | + 3.74                  | + 20.93                 | + 0.001                                    | + 0.003 | + 0.012                                | + 0.012 | + 3.81                                  | + 3.32    | + 2.11       | + 4.33     | + 2.31       |
| Busing Program                       | + 16.92                 | + 38.67                 | + 0.008                                    | + 0.009 | + 0.039                                | + 0.036 | + 3.70                                  | + 4.11    | + 3.98       | + 11.72    | + 4.72       |
| Decentralized                        | -4.86                   | + 27.48                 | + 0.002                                    | + 0.002 | + 0.005                                | + 0.004 | + 9.30                                  | + 5.18    | + 2.26       | + 7.05     | + 2.78       |
| Mandatory App.                       | + 65.62                 | + 43.40                 | + 0.018                                    | + 0.017 | + 0.113                                | + 0.104 | -14.45                                  | + 2.94    | + 4.96       | + 15.94    | + 5.26       |
| Mandatory App. + Busing              | + 83.02                 | + 45.55                 | + 0.022                                    | + 0.020 | + 0.136                                | + 0.125 | -15.05                                  | + 1.94    | + 5.31       | + 18.93    | + 5.64       |
| Mandatory App. + Broad Info          | + 66.26                 | + 43.63                 | + 0.018                                    | + 0.017 | + 0.114                                | + 0.104 | -14.44                                  | + 3.02    | + 4.95       | + 16.18    | + 5.29       |
| Mandatory App. + Broad Info + Busing | + 83.08                 | + 45.55                 | + 0.022                                    | + 0.021 | + 0.136                                | + 0.125 | -14.98                                  | + 1.95    | + 5.28       | + 18.94    | + 5.61       |
| Maximized Achievement                | —                       | + 45.83                 | + 0.038                                    | + 0.033 | + 0.266                                | + 0.240 | -17.10                                  | + 11.62   | + 4.98       | + 19.36    | + 4.91       |

*Note:* This table reports simulated results from a model of application, enrollment, and achievement, with modifications to the achievement model demand models relative to our preferred model in the main body of the paper. In particular, on the demand side we augment Auxiliary Model I with a busing eligibility demand shifter. On the outcome side, we allow for distance treatment effect heterogeneity, relaxing the exclusion restriction in the main body of the paper and relying on the busing eligibility shifter and lottery offers for identification. The corresponding outcome model estimates are reported in Appendix Table A.11. Panel (a) reports results for the baseline opt-in system. Panel (b) reports counterfactual impacts, expressed as the difference relative to the baseline scenario. We report the percent of students that apply to choice schools as their most-preferred option (Column 1), and the percent of choice school seats that are filled (Column 2). For achievement, we report district-wide achievement (Columns 3 and 4) and treatment on the treated estimates (Columns 5 and 6). Demographic outcomes include the percent of students within each racial or income-group enrolling in a choice school (Columns 7-11). The General Information scenario increases the  $\theta_i$  of a randomly selected 50% of students in the district, while the Targeted Information scenario increases the  $\theta_i$  of a randomly selected 50% of students who are enrolled in schools whose average achievement is below median. The No Travel Costs scenario mirrors a generous busing policy that sets travel costs to zero. The Decentralized scenario allows families to submit multiple independent applications to schools, which then run their own lotteries, potentially generating multiple simultaneous offers. The Mandatory scenario eliminates application costs, allowing families to rank their neighborhood school as their most-preferred option and assignments are done via a deferred acceptance mechanism. For each counterfactual, we simulate the economy 100 times. For scenarios that retain school-run lotteries (information, busing, and decentralized choice), we calculate equilibrium best response admissions probabilities that schools would set under the changed demand implied by the policy (see Appendix Section C for details).

## APPENDIX B: DATA APPENDIX

### B.1. *Details of the Original Data Collection*

We collected data for the 150 highest enrollment school districts in the United States that cover roughly 27% of public school enrollment nationwide. The data we collected was obtained through both online research and calls to district administrators made by research assistants. For each school district, we were interested in obtaining measures of the availability of intra-district choice, the assignment process, the timing, participation details, the type of choice offerings a district provided, and an assessment of the difficulty in obtaining this information via the web and phone. The process began with a research assistant examining the district's webpage and collecting all relevant information that was online. Next, research assistants made calls to school districts to obtain information via conversations with administrators. Most school districts were researched by three research assistants. In the case of discrepancies, the principal investigators verified the accuracy of the information.

### B.2. *List of Districts*

Appendix Table [B.1](#) lists the surveyed districts and classifies their choice systems as mandatory (M), opt-in (O), or decentralized (D), along with each district's state, enrollment, and national enrollment rank. These classifications define the groups shown in Figure 1 of the main paper.

TABLE B.1  
CHOICE-SYSTEM CLASSIFICATIONS IN THE LARGEST U.S. SCHOOL DISTRICTS

| Ranks 1–50 |                             |              | Ranks 51–100 |                                   |              | Ranks 101–150 |                                  |              |
|------------|-----------------------------|--------------|--------------|-----------------------------------|--------------|---------------|----------------------------------|--------------|
| Rank       | District (State)            | Enroll. Sys. | Rank         | District (State)                  | Enroll. Sys. | Rank          | District (State)                 | Enroll. Sys. |
| 1          | New York City (NY)          | 845,509 M    | 51           | Conroe ISD (TX)                   | 72,352 D     | 101           | Piano ISD (TX)                   | 47,899 D     |
| 2          | Los Angeles Unified (CA)    | 419,929 M    | 52           | Fort Worth ISD (TX)               | 71,060 D     | 102           | Frederick County (MD)            | 47,681 D     |
| 3          | Miami-Dade (FL)             | 335,500 O    | 53           | Fresno Unified (CA)               | 68,568 O     | 103           | Pasadena ISD (TX)                | 47,486 O     |
| 4          | Chicago Public Schools (IL) | 322,809 O    | 54           | Guilford County Schools (NC)      | 67,832 D     | 104           | Socorro ISD (TX)                 | 47,304 D     |
| 5          | Clark County (NV)           | 304,565 O    | 55           | Milwaukee (WI)                    | 66,864 M     | 105           | St. Lucie (FL)                   | 46,987 O     |
| 6          | Broward (FL)                | 251,408 O    | 56           | Seminole (FL)                     | 65,443 O     | 106           | Wichita (KS)                     | 46,516 O     |
| 7          | Hillsborough (FL)           | 224,152 O    | 57           | Virginia Beach City (VA)          | 64,896 D     | 107           | Round Rock ISD (TX)              | 46,197 O     |
| 8          | Orange (FL)                 | 206,815 O    | 58           | Chesapeake County (VA)            | 64,132 D     | 108           | Hamilton County (TN)             | 45,790 O     |
| 9          | Palm Beach (FL)             | 189,777 O    | 59           | Chesterfield County (VA)          | 63,962 M     | 109           | Boston (MA)                      | 45,742 M     |
| 10         | Houston ISD (TX)            | 184,109 M    | 60           | Long Beach Unified (CA)           | 63,777 O     | 110           | Marion (FL)                      | 45,547 O     |
| 11         | Gwinnett County (GA)        | 182,214 D    | 61           | Washoe County (NV)                | 62,742 O     | 111           | Sarasota (FL)                    | 45,314 O     |
| 12         | Fairfax County (VA)         | 180,714 D    | 62           | Volusia (FL)                      | 62,603 O     | 112           | Columbus City Schools (OH)       | 45,181 O     |
| 13         | Fairfax County Schools (NC) | 161,481 O    | 63           | Elk Grove Unified (CA)            | 62,603 O     | 113           | San Bernardino City Unified (CA) | 44,712 O     |
| 14         | Wake County Schools (NC)    | 160,223 D    | 64           | Douglas County RE-1 (CO)          | 61,409 O     | 114           | San Antonio ISD (TX)             | 44,670 O     |
| 15         | Montgomery County (MD)      | 144,116 O    | 65           | Knox County (TN)                  | 60,604 O     | 115           | Nebo District (UT)               | 44,446 O     |
| 16         | Charlotte-Mecklenburg (NC)  | 143,196 D    | 66           | Granite District (UT)             | 60,270 O     | 116           | San Antonio ISD (TX)             | 44,039 O     |
| 17         | Dallas ISD (TX)             | 131,310 D    | 67           | Jordan District (UT)              | 59,145 O     | 117           | Portland SD 11 (OR)              | 43,864 D     |
| 18         | Prince George's County (MD) | 131,310 D    | 68           | Aldine ISD (TX)                   | 57,844 D     | 118           | Killeen ISD (TX)                 | 43,563 D     |
| 19         | Duval (FL)                  | 127,971 O    | 69           | Howard County (MD)                | 57,643 D     | 119           | Anchorage (AK)                   | 43,291 O     |
| 20         | Cypress-Fairbanks ISD (TX)  | 118,470 O    | 70           | North East ISD (TX)               | 57,343 D     | 120           | Clovis Unified (CA)              | 43,258 O     |
| 21         | Philadelphia City SD (PA)   | 117,907 O    | 71           | Mesa Unified District (4235) (AZ) | 57,263 D     | 121           | Henry County (GA)                | 43,116 O     |
| 22         | Polk (FL)                   | 111,041 O    | 72           | Forsyth County (GA)               | 54,984 D     | 122           | Kern High (CA)                   | 42,832 O     |
| 23         | Baltimore County (MD)       | 110,215 D    | 73           | Arlington ISD (TX)                | 54,750 O     | 123           | Chandler Unified No. 80 (AZ)     | 42,791 O     |
| 24         | Memphis-Shelby County (TN)  | 105,368 O    | 74           | Klein ISD (TX)                    | 53,093 O     | 124           | Newark Public Schools (NJ)       | 42,593 D     |
| 25         | Cobb County (GA)            | 101,095 O    | 75           | Manatee (FL)                      | 52,157 O     | 125           | Leander ISD (TX)                 | 42,509 D     |
| 26         | Northside ISD (TX)          | 100,064 M    | 76           | Winston-Salem/Forsyth (NC)        | 51,808 O     | 126           | East Baton Rouge Parish (LA)     | 42,016 O     |
| 27         | Lee (FL)                    | 95,492 O     | 77           | Cherry Creek No. 5 (CO)           | 51,693 O     | 127           | Cherokee County (GA)             | 41,909 D     |
| 28         | San Diego Unified (CA)      | 94,793 M     | 78           | Omaha Public Schools (NE)         | 51,659 M     | 128           | Williamson County (TN)           | 41,487 D     |
| 29         | Jefferson County (KY)       | 94,785 D     | 79           | Garland ISD (TX)                  | 51,595 O     | 129           | Union County (NC)                | 41,654 D     |
| 30         | Katy ISD (TX)               | 91,938 O     | 80           | Rutherford County (TN)            | 50,929 D     | 130           | Lincoln Public Schools (NE)      | 41,203 D     |
| 31         | DeKalb County (GA)          | 91,021 O     | 81           | St. Johns (FL)                    | 50,831 O     | 131           | United ISD (TX)                  | 40,929 O     |
| 32         | Pinellas (FL)               | 88,258 M     | 82           | Mobile County (AL)                | 50,831 O     | 132           | Tucson Unified (AZ)              | 40,836 O     |
| 33         | Prince William County (VA)  | 88,043 O     | 83           | DC Public Schools (DC)            | 50,770 O     | 133           | Capistrano Unified (CA)          | 40,576 D     |
| 34         | Denver (CO)                 | 87,051 D     | 84           | Clayton County (GA)               | 50,467 O     | 134           | Chesapeake City (VA)             | 40,200 D     |
| 35         | Fulton County (GA)          | 85,855 D     | 85           | Seattle Public Schools (WA)       | 49,864 D     | 135           | Clear Creek ISD (TX)             | 39,474 O     |
| 36         | Alpine District (UT)        | 84,336 D     | 86           | Jefferson Parish (LA)             | 49,660 O     | 136           | Alief ISD (TX)                   | 39,362 O     |
| 37         | Pasco (FL)                  | 81,636 D     | 87           | Hemlock County (VA)               | 49,314 D     | 137           | Clay (FL)                        | 39,345 O     |
| 38         | Anne Arundel County (MD)    | 80,694 D     | 88           | Charleston 01 (SC)                | 49,139 D     | 138           | Montgomery County (TN)           | 39,265 D     |
| 39         | Loudoun County (VA)         | 80,468 O     | 89           | Corona-Norco Unified (CA)         | 48,736 M     | 139           | Berkley 01 (SC)                  | 38,991 D     |
| 40         | Fort Bend ISD (TX)          | 76,756 D     | 90           | Atlanta Public Schools (GA)       | 48,552 D     | 140           | Joint SD No. 2 (ID)              | 38,855 D     |
| 41         | Davidson County (TN)        | 74,289 O     | 91           | El Paso ISD (TX)                  | 48,440 D     | 141           | Riverside Unified (CA)           | 38,734 D     |
| 42         | Greenville 01 (SC)          | 73,810 M     | 92           | Cumberland County Schools (NC)    | 48,285 O     | 142           | St. Tammany Parish (LA)          | 38,631 D     |
| 43         | Albuquerque (NM)            | 72,830 M     | 93           | San Francisco Unified (CA)        | 48,271 O     | 143           | Anoka-Hennepin (MN)              | 38,488 O     |
| 44         | Baltimore City (MD)         | 72,499 O     | 94           | Humble ISD (TX)                   | 48,205 O     | 144           | San Juan Unified (CA)            | 38,268 O     |
| 45         | Osceola (FL)                | 72,499 O     | 95           | Lewisville ISD (TX)               | 48,205 O     | 145           | Sacramento City Unified (CA)     | 38,265 O     |
| 46         | Brevard (FL)                | 72,499 O     | 96           | Lake (FL)                         | 48,205 O     | 146           | Mesquite ISD (TX)                | 38,178 D     |
| 47         | Jefferson County R-1 (CO)   | 72,499 O     | 97           | Collier (FL)                      | 48,205 O     | 147           | Aurora Joint No. 28 (CO)         | 38,158 D     |
| 48         | Austin ISD (TX)             | 72,499 O     | 98           | Detroit Public Schools (MI)       | 48,205 O     | 148           | Harford County (MD)              | 38,158 D     |
| 49         | Davis District (UT)         | 72,499 O     | 99           | Horry 01 (SC)                     | 48,205 O     | 149           |                                  |              |
| 50         |                             |              | 100          |                                   |              | 150           |                                  |              |

Note: Districts are ordered by national enrollment rank. Enrollment systems are classified as mandatory (M), opt-in (O), or decentralized (D). State names are abbreviated using standard two-letter postal abbreviations. District names are shortened where necessary to remove administrative boilerplate; the accompanying source file retains the names as originally provided. The source lists contain 143 districts. Supplied ranks are preserved exactly, including two districts reported at Rank 40 and the absence of Ranks 13, 41, 43, 56, 64, 115, 129, and 143.

## APPENDIX C: COUNTERFACTUAL DETAILS

This appendix section provides details for our counterfactual analysis. Each counterfactual policy exercise involves simulating environments using our estimated demand and potential outcome models. The outcome model estimates are based on the linear control function approach in Equation (11) of the main paper. All simulations assume the supply of choice schools is held fixed to the observed years included in our sample (2004-2013). The specific alterations to the environment allow us to isolate the policy effect of interest relative to the existing opt-in (voluntary) system. For each counterfactual, we simulate  $R = 100$  economies and report averages of statistics across simulations. We discuss the details for each counterfactual next.

### C.1. *Information Interventions*

As mentioned in the main body of the paper, information interventions involve boosting a student's  $\theta_i$  by one standard deviation of their estimated type-specific distribution. Importantly, we only allow the shift in  $\theta_i$  to affect application and enrollment decisions but use the original  $\theta_i$  when calculating counterfactual potential outcomes.

Because changes to  $\theta_i$  lead to changes in application rates relative to the baseline scenario, we must calculate equilibrium admission probabilities that students consider when making application decisions to choice schools. We follow the logic of Walters (2018) and Avery et al. (2025) to solve for equilibrium admission probabilities under the new environment. For each counterfactual simulation,  $r = 1, \dots, R$ , we solve for the equilibrium admission probabilities given the draw of idiosyncratic preference ( $\theta_i$ ) as well as their post-lottery ( $\xi_{ij}$  and cost shocks ( $\eta_i$ ) that govern application and enrollment decisions. Once we solve for the equilibrium  $\pi_{ij}^{BR}$  for each school  $j$ , we then calculate the resulting statistics for the counterfactual simulation. We repeat this  $R = 100$  times and average statistics across simulation rounds.

### C.2. *Busing Program*

The busing policies that we consider are approximated in our model as an elimination of travel costs. As in the information interventions, this leads to a change in demand, so we must estimate equilibrium admission probabilities in this new environment. We follow the same procedure outlined above.

### C.3. *Decentralized*

The decentralized market allows families to submit applications to multiple schools and potentially receive multiple offers, in contrast to our baseline model, which restricts households to a single application to mirror LAUSD policy during the sample period. To implement this counterfactual, we modify how application vectors are constructed. A key change is to augment the application cost structure by introducing marginal costs: whereas the baseline model includes only a fixed application cost, the decentralized setting requires a per-application marginal cost to prevent students from submitting excessively large portfolios. We follow the prior literature and calibrate this marginal cost to match the fixed-to-marginal cost ratio in Avery et al. (2025). In this environment, students form application portfolios by weighing the fixed cost of initiating the application process together with a constant marginal cost for each additional school they choose to include.

Because fixed application costs introduce non-convexities, we cannot directly apply the off-the-shelf marginal improvement algorithm from Chade and Smith (2006). To address this, we use a modified procedure: we first “seed” the algorithm by starting with the single school

that delivers the highest stand-alone expected utility, and then apply the standard Chade-Smith marginal-improvement rule to consider adding additional schools to their portfolio. At the end of the process, we compare the expected utility from the resulting application portfolio to the utility from not applying to any school.

To verify that this modified procedure produces the optimal application portfolio, we conduct Monte Carlo simulations comparing the algorithm’s output to brute-force enumeration of all feasible portfolios for a representative sample of students. In all cases, the modified greedy algorithm recovered the same optimal portfolio as the exhaustive search, confirming that the procedure correctly identifies the utility-maximizing application vector. As in our other counterfactuals, we then compute equilibrium admission probabilities implied by the demand induced by the transition to a decentralized market.

#### C.4. *Mandatory Application*

The mandatory application policy involves an application mandate and the introduction of a deferred acceptance mechanism to govern assignments. Importantly, we allow families to rank multiple schools. In practice, this means we eliminate application costs and require families to rank schools in order of preferences governed by the estimated demand parameters and their observables. Our use of an uncapped choice school list avoids having to estimate equilibrium admission probabilities. Therefore, we simulate  $R = 100$  environments and average statistics across simulation rounds. We assume students enroll in their assigned school.

#### C.5. *Mandatory Application and Complementary Policies*

For counterfactuals that combine the mandatory application policy with other complementary reforms, we follow the procedures discussed above. Because each involves the use of a deferred acceptance mechanism with unrestricted list lengths, we do not estimate equilibrium admission probabilities. We assume students enroll in their assigned school.

#### REFERENCES

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